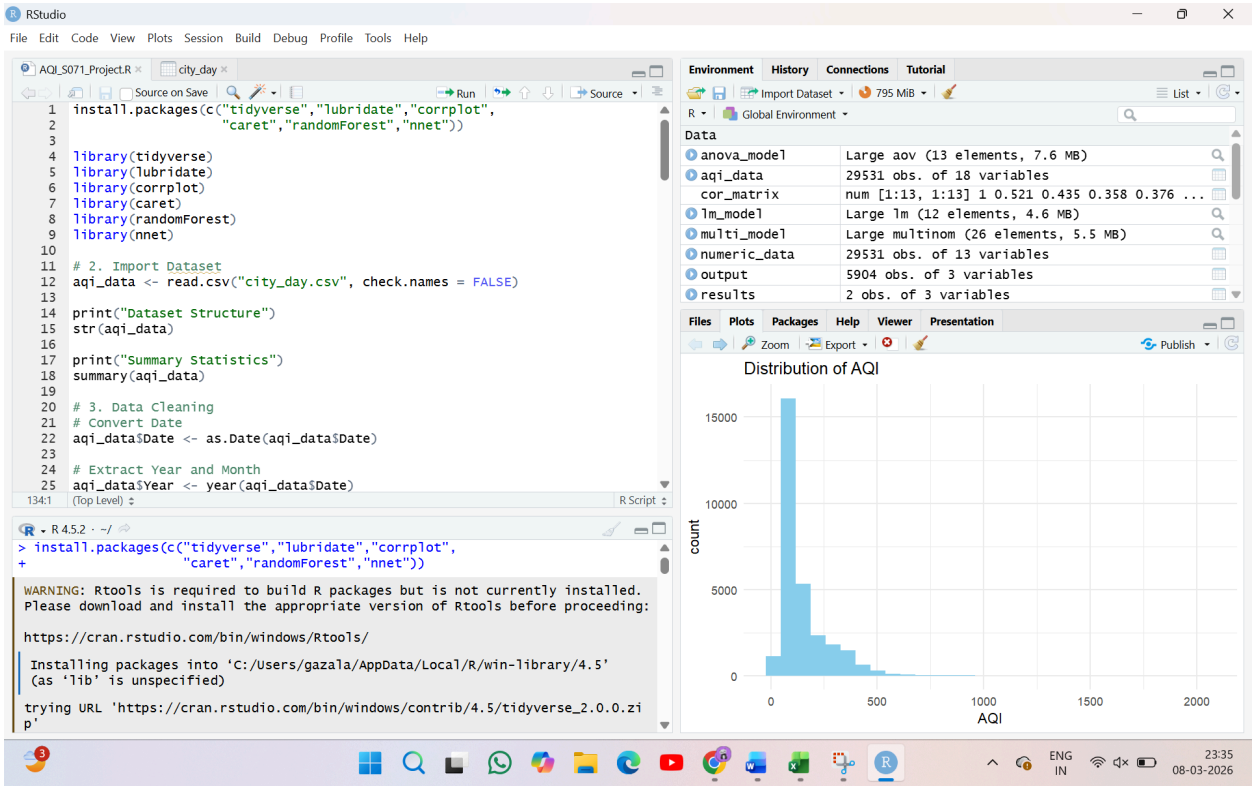
SHETH L.U.J AND SIR M.V COLLEGE

Subject: Data Analysis



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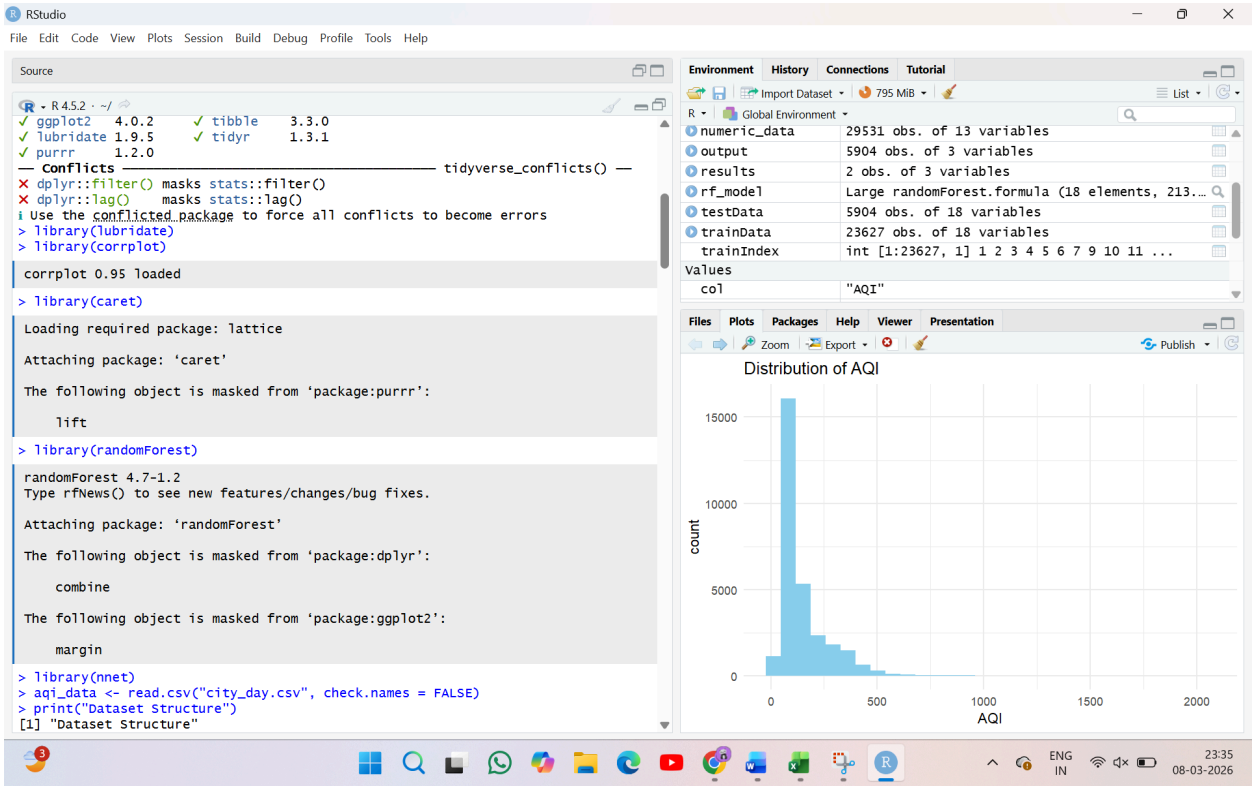

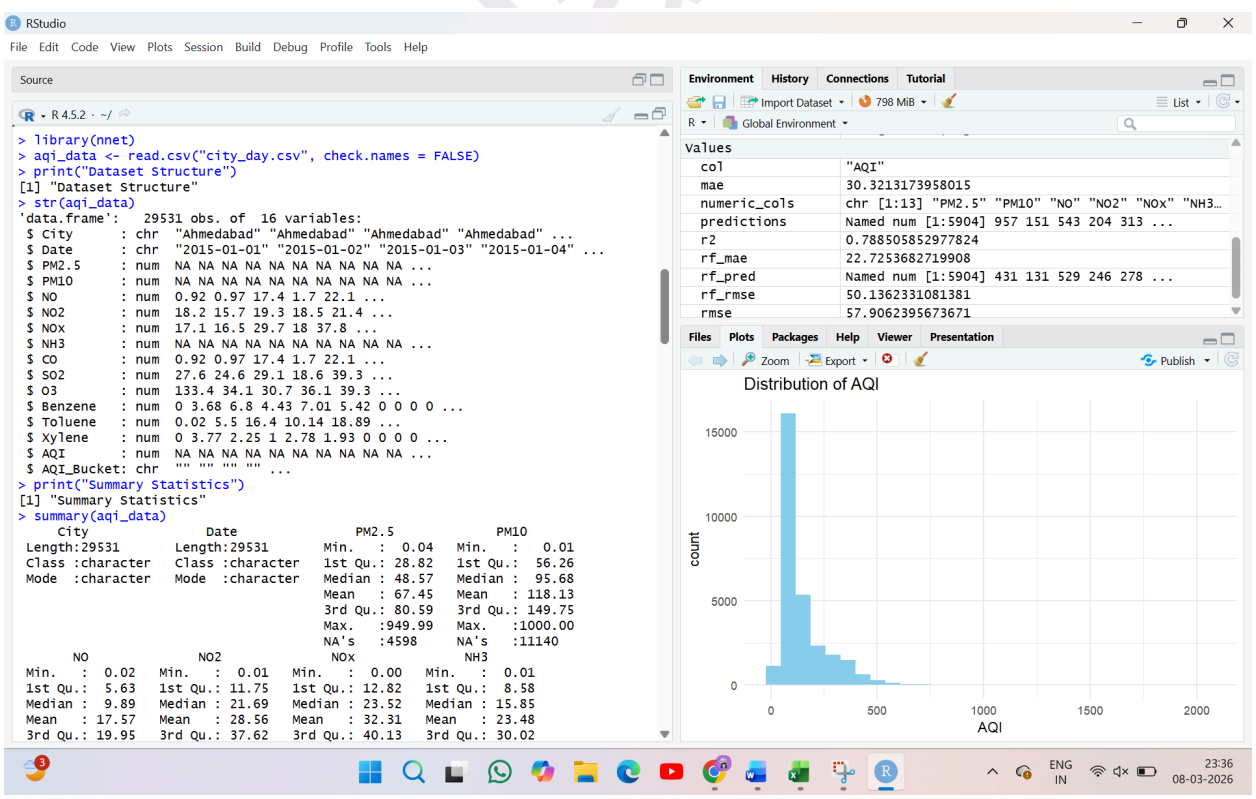

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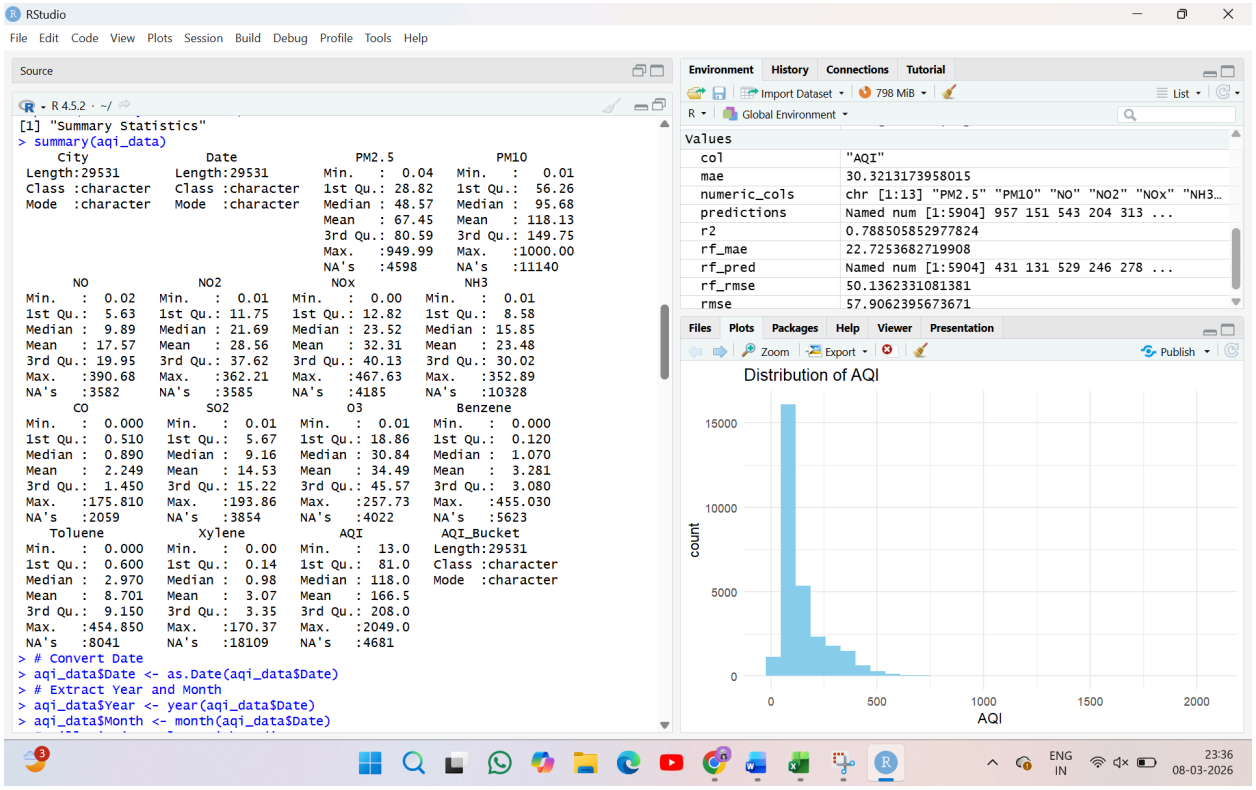

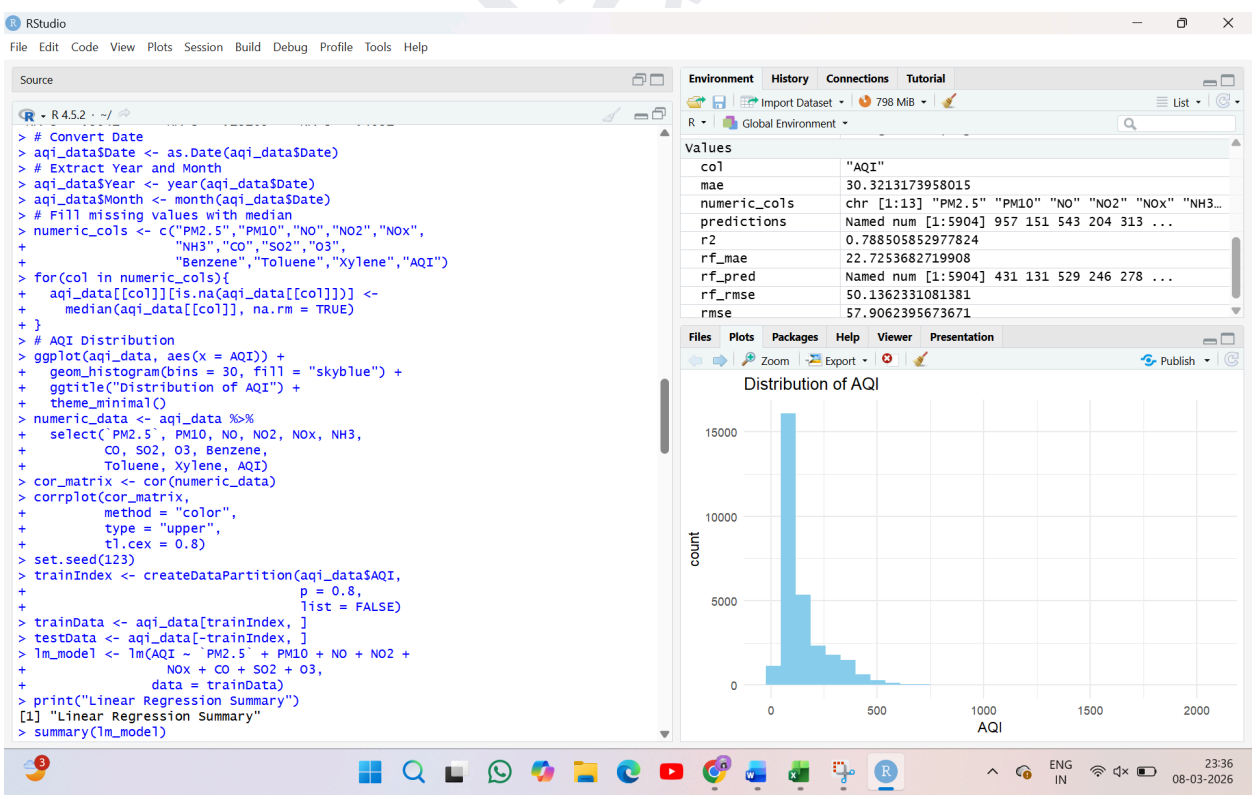

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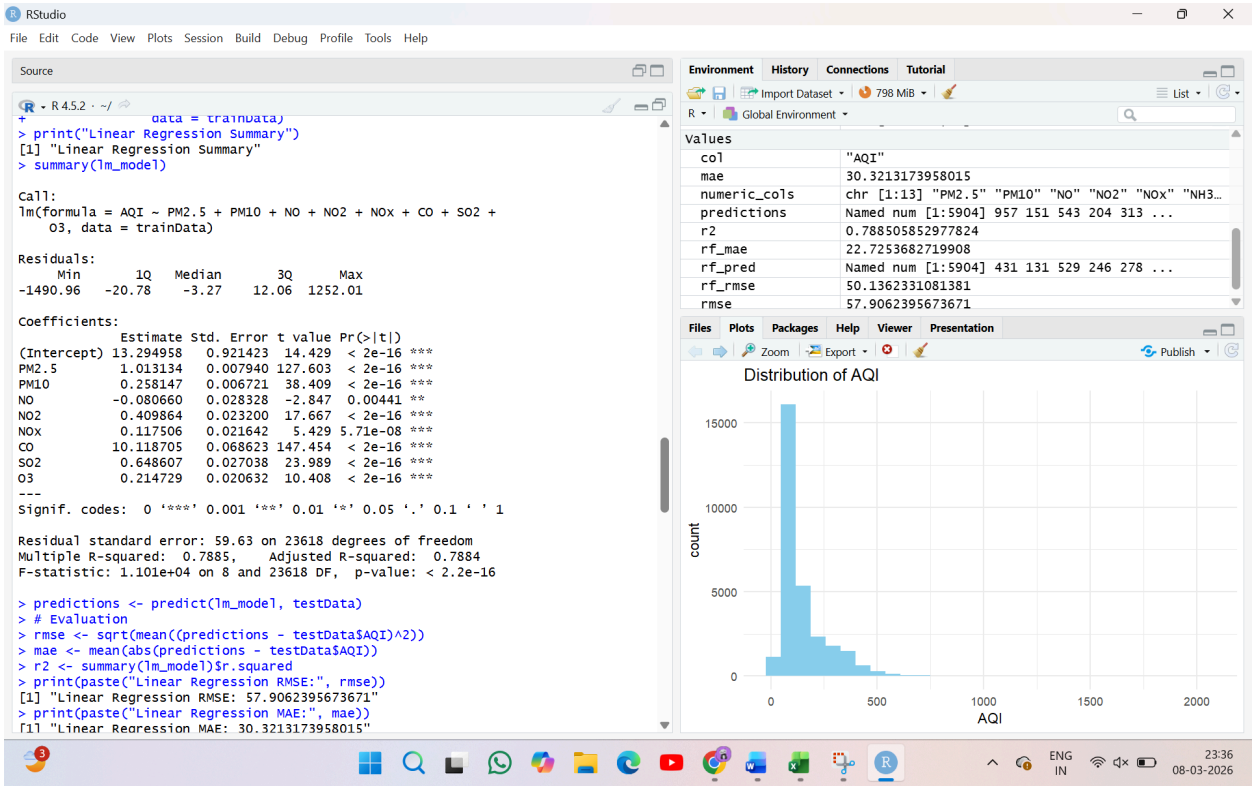
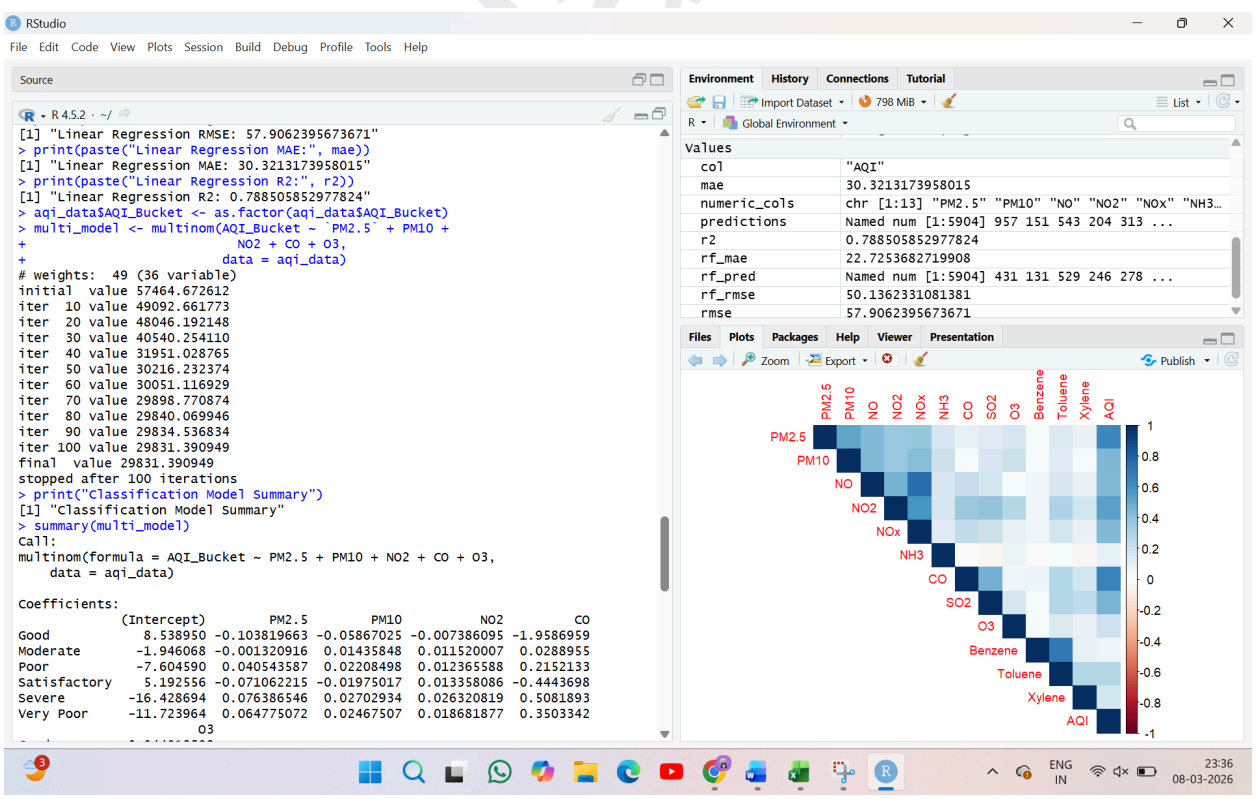


```
+                 data = trainData)
> print("Linear Regression Summary")
[1] "Linear Regression Summary"
> summary(lm_model)

Call:
lm(formula = AQI ~ PM2.5 + PM10 + NO + NO2 + NOx + CO + SO2 +
    O3, data = trainData)

Residuals:
     Min       1Q   Median       3Q      Max
-1490.96   -20.78    -3.27    12.06  1252.01

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.294958   0.921423  14.429  < 2e-16 ***
PM2.5        1.013134   0.007940 127.603  < 2e-16 ***
PM10         0.258147   0.006721  38.409  < 2e-16 ***
NO          -0.080660   0.028328  -2.847  0.00441 **
NO2          0.409864   0.023200  17.667  < 2e-16 ***
NOx          0.117506   0.021642   5.429 5.71e-08 ***
CO          10.118705   0.068623 147.454  < 2e-16 ***
SO2          0.648607   0.027038  23.989  < 2e-16 ***
O3           0.214729   0.020632  10.408  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 59.63 on 23618 degrees of freedom
Multiple R-squared:  0.7885,    Adjusted R-squared:  0.7884
F-statistic: 1.101e+04 on 8 and 23618 DF,  p-value: < 2.2e-16

> predictions <- predict(lm_model, testData)
> # Evaluation
> rmse <- sqrt(mean((predictions - testData$AQI)^2))
> mae <- mean(abs(predictions - testData$AQI))
> r2 <- summary(lm_model)$r.squared
> print(paste("Linear Regression RMSE:", rmse))
[1] "Linear Regression RMSE: 57.9062395673671"
> print(paste("Linear Regression MAE:", mae))
[1] "Linear Regression MAE: 30.3213173958015"
```

Values	
col	"AQI"
mae	30.3213173958015
numeric_cols	chr [1:13] "PM2.5" "PM10" "NO" "NO2" "NOx" "NH3…
predictions	Named num [1:5904] 957 151 543 204 313 ...
r2	0.788505852977824
rf_mae	22.7253682719908
rf_pred	Named num [1:5904] 431 131 529 246 278 ...
rf_rmse	50.1362331081381
rmse	57.9062395673671



```
[1] "Linear Regression RMSE: 57.9062395673671"
> print(paste("Linear Regression MAE:", mae))
[1] "Linear Regression MAE: 30.3213173958015"
> print(paste("Linear Regression R2:", r2))
[1] "Linear Regression R2: 0.788505852977824"
> aqi_data$AQI_Bucket <- as.factor(aqi_data$AQI_Bucket)
> multi_model <- multinom(AQI_Bucket ~ `PM2.5` + PM10 +
+                         NO2 + CO + O3,
+                       data = aqi_data)
# weights:  49 (36 variable)
initial  value 57464.672612
iter  10 value 49092.661773
iter  20 value 48046.192148
iter  30 value 40540.254110
iter  40 value 31951.028765
iter  50 value 30216.232374
iter  60 value 30051.116929
iter  70 value 29898.770874
iter  80 value 29840.069946
iter  90 value 29834.536834
iter 100 value 29831.390949
final  value 29831.390949
stopped after 100 iterations
> print("Classification Model Summary")
[1] "Classification Model Summary"
> summary(multi_model)
Call:
multinom(formula = AQI_Bucket ~ PM2.5 + PM10 + NO2 + CO + O3,
    data = aqi_data)

Coefficients:
             (Intercept)        PM2.5        PM10          NO2          CO
Good            8.538950 -0.103819663 -0.05867025 -0.007386095 -1.9586959
Moderate       -1.946068 -0.001320916  0.01435848  0.011520007  0.0288955
Poor           -7.604590  0.040543587  0.02208498  0.012365588  0.2152133
Satisfactory    5.192556 -0.071062215 -0.01975017  0.013358086 -0.4443698
Severe        -16.428694  0.076386546  0.02702934  0.026320819  0.5081893
Very Poor     -11.723964  0.064775072  0.02467507  0.018681877  0.3503342
                     O3
```

Values	
col	"AQI"
mae	30.3213173958015
numeric_cols	chr [1:13] "PM2.5" "PM10" "NO" "NO2" "NOx" "NH3…
predictions	Named num [1:5904] 957 151 543 204 313 ...
r2	0.788505852977824
rf_mae	22.7253682719908
rf_pred	Named num [1:5904] 431 131 529 246 278 ...
rf_rmse	50.1362331081381
rmse	57.9062395673671

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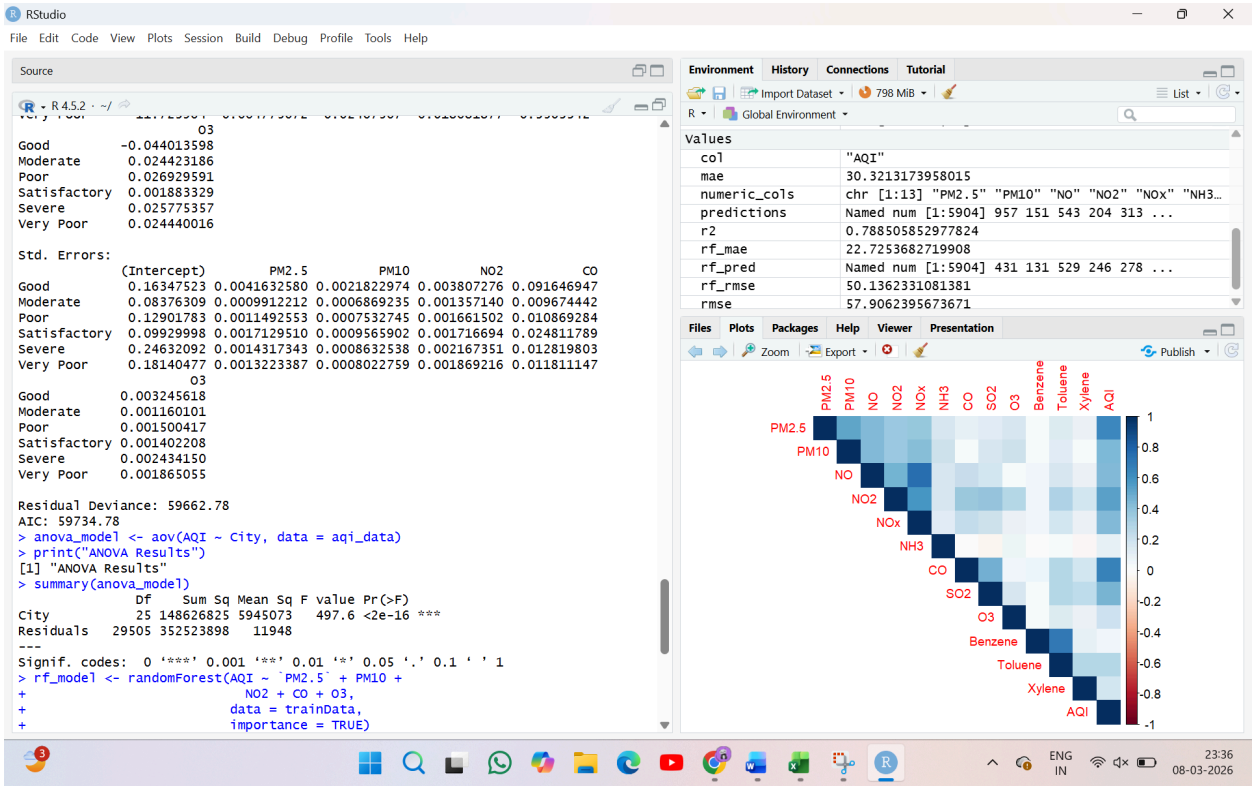
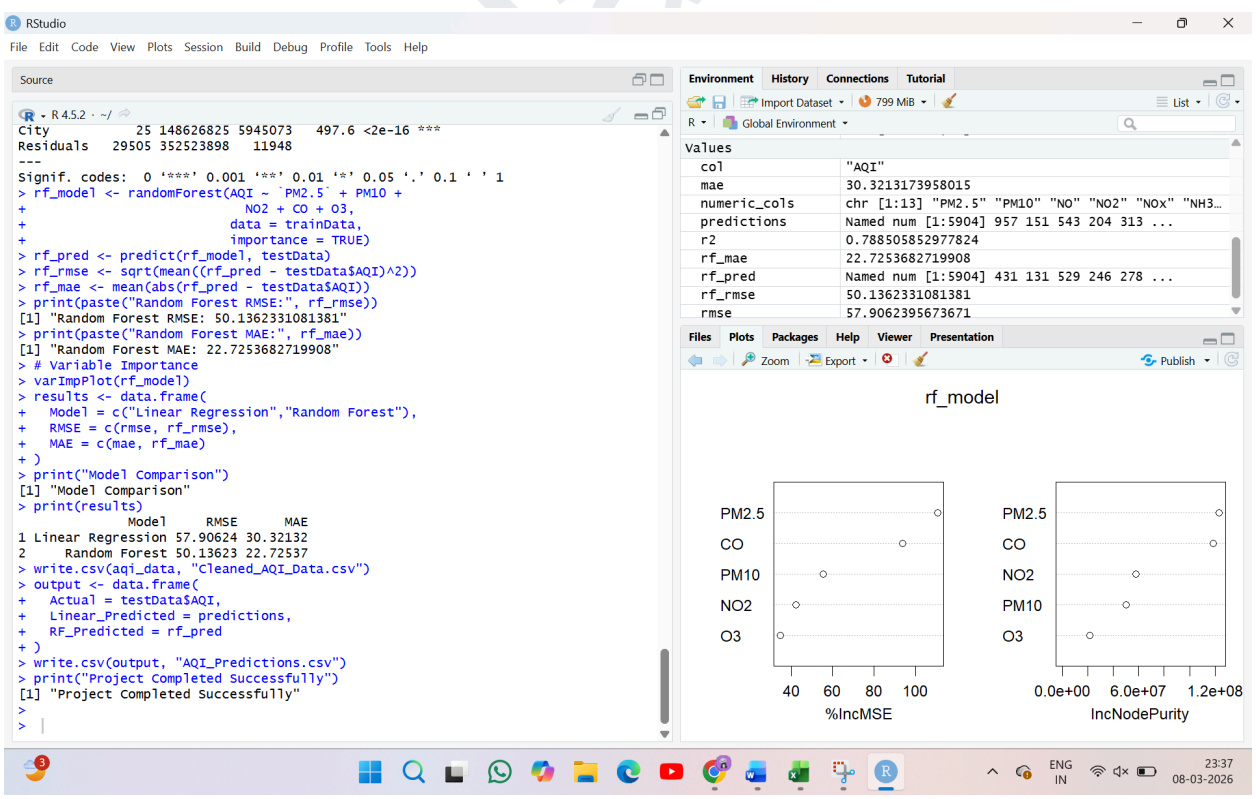
Subject: Data Analysis





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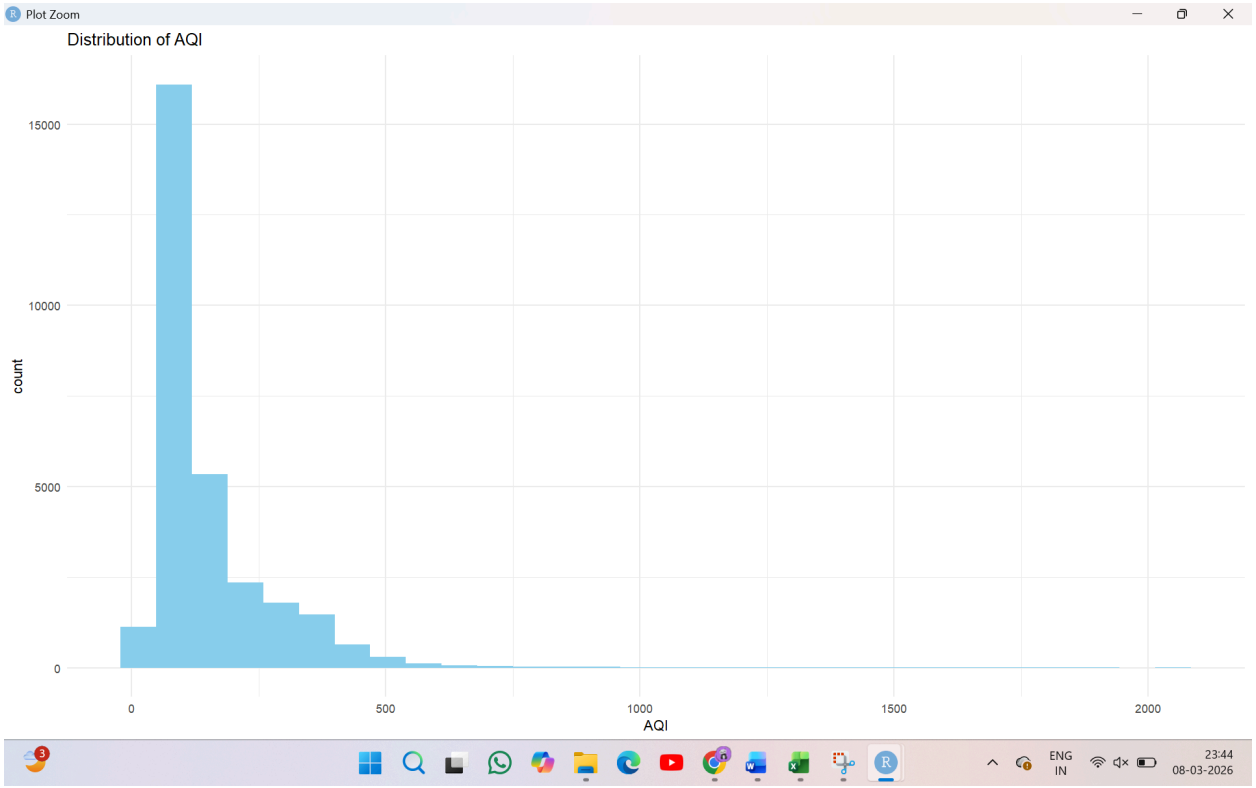

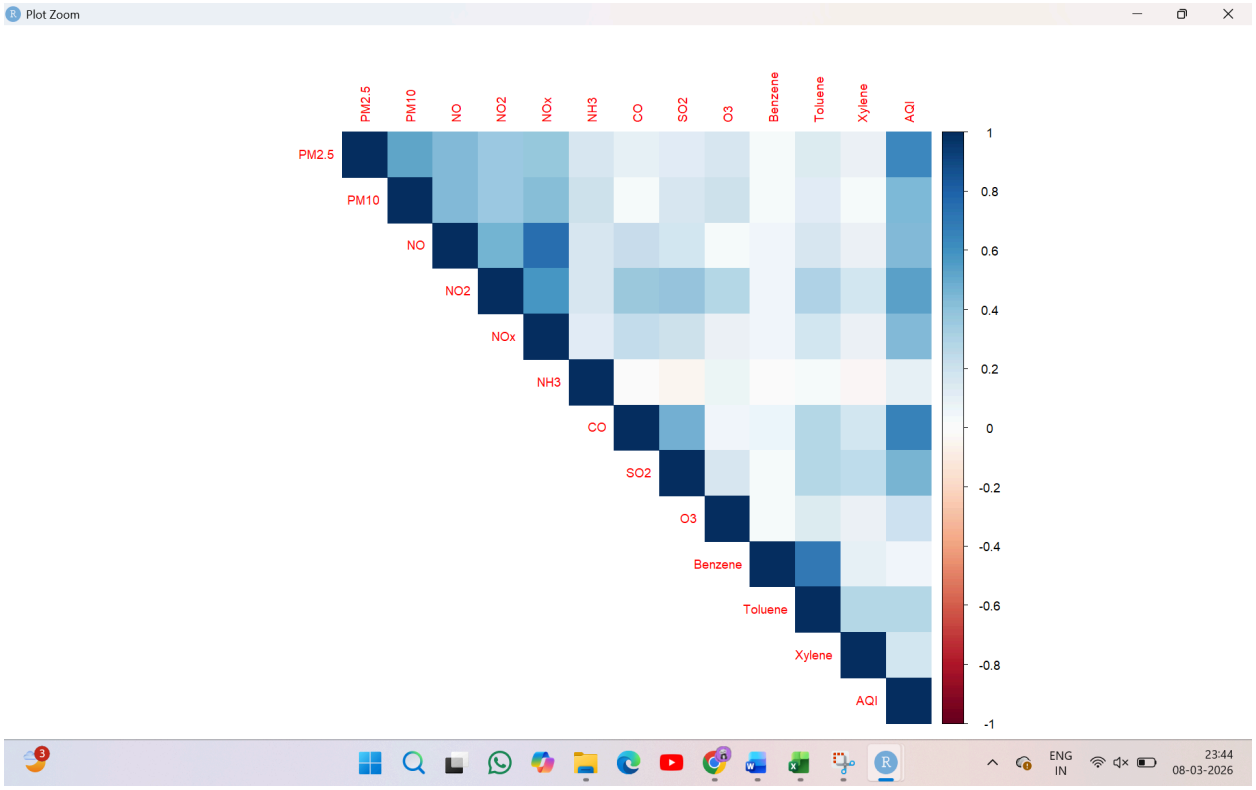

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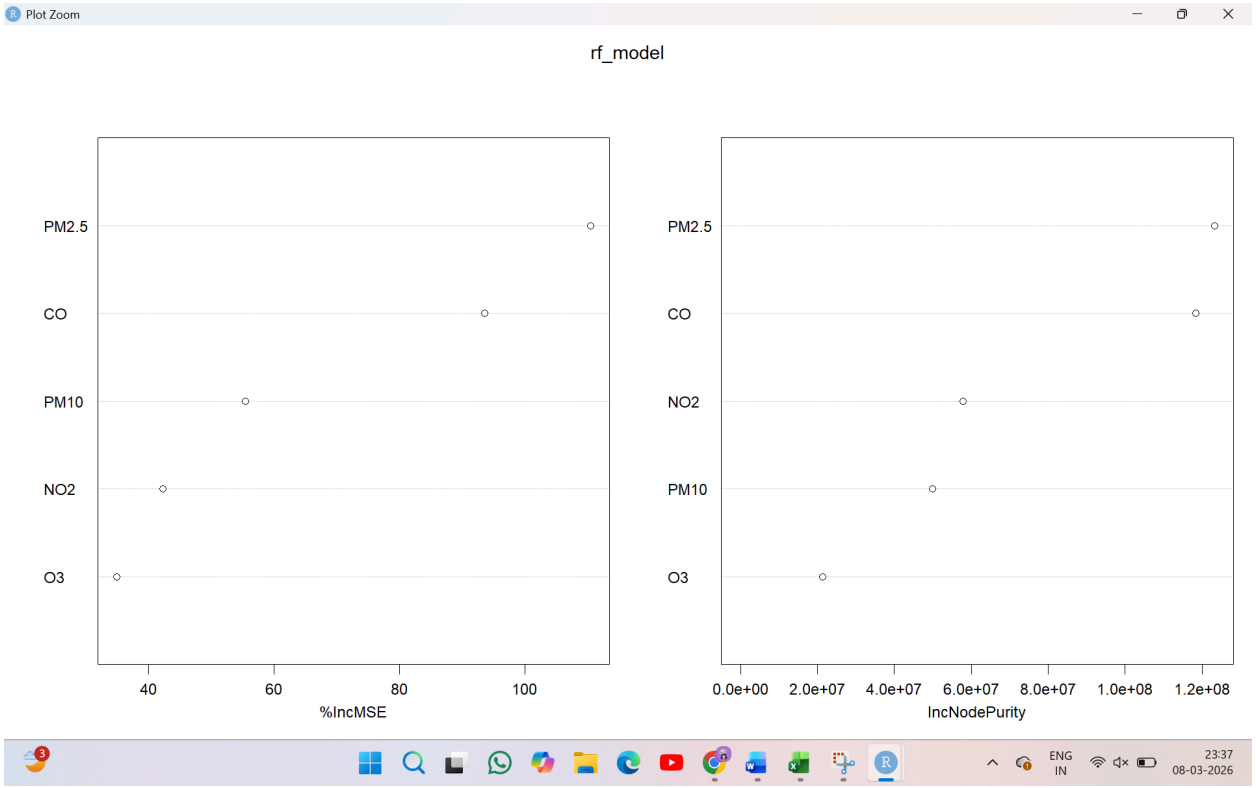

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